A Knowledge-Enhanced Chatbot System Using Verified Review Intelligence for E-commerce platforms

UIT2718-PROJECT WORK-PHASE 2

A PROJECT REPORT

Submitted by

Lokesh Adithya B	3122215002053
Rupam Banik	3122215002086
D Sachin Abhishek	3122215002088

SSN COLLEGE OF ENGINEERING, KALAVAKKAM

APRIL 2024

Sri Siva Subramaniya Nadar College of Engineering
(An Autonomous Institution, Affiliated to Anna University)

Sri Sivasubramaniya Nadar College of Engineering

(An Autonomous Institution, Affiliated to Anna University)

Department of Information Technology

BONAFIDE CERTIFICATE

Certified that this report, titled "**A Knowledge-Enhanced Chatbot System Using Verified Review Intelligence for E-commerce platforms**" is the bonafide work of **Lokesh Adithya B (3122215002053), D Sachin Abhishek (3122215002088) and Rupam Banik (3122215002086)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Dr. A. Shahina
Professor and Head of Department
Dept of Information Technology
SSN College of Engineering
Kalavakkam – 603 110

Dr. E. Suganya
Assisstant Professor
Dept of Information Technology
SSN College of Engineering
Kalavakkam – 603 110

Submitted for project viva-voce examination held on 08th APRIL 2024

EXTERNAL EXAMINER INTERNAL EXAMINER

ABSTRACT

Customers generally find it hard to differentiate between real and fake reviews of products because e-commerce has grown rapidly. Shopping choices can be significantly affected by such difficulty, and hence an authentic review verification system is needed. In a bid to determine and mark off fake reviews, our project introduces an advanced review verifying mechanism powered by machine learning and natural language processing (NLP). Our approach assures greater transparency and trust in e-commerce product reviews using authentic review intelligence.

With the Phi-3 enhanced model, we have also developed an intelligent chatbot that assists users in finding product reviews more effectively. To give users contextual responses to queries regarding the performance, usability, and durability of products, the chatbot extracts, summarizes, and analyzes verified reviews. By eliminating the necessity for users to sift through numerous reviews manually, this automated review analysis speeds up the decision-making process.

By means of real-time learning mechanisms and user feedback loops, our system constantly adapts to evolving deceptive methods, ensuring long-term effectiveness. We enhance the user experience and provide a simple-to-use interface for interaction with verified product reviews by incorporating a robust frontend application. Both customers and ethical sellers benefit from this solution's creation of a more trustworthy e-commerce environment.

ACKNOWLEDGEMENT

I thank **ALMIGHTY GOD** who gave me the wisdom to complete this Project. My sincere thanks to our beloved founder **Dr. SHIV NADAR, Chairman, HCL Technologies**. I also express my sincere thanks to **Ms. KALA VIJAYAKUMAR**, President, SSN Institution and our Principal **Dr. V.E. ANNAMALAI**, for all the help he has rendered during this course of study.

We are highly indebted to **Dr. A. SHAHINA, Head of the Department** for providing us with the opportunity and facilities to take up this project.

I am deeply obliged and indebted to the timeless help and guidance provided by **Dr. E. SUGANYA, Assisstant Professor**, Department of Information Technology and also express my heartfelt thanks for making this project a great success.

I also thank all the faculty of the Department of Information Technology for their kind advice, support and encouragement and last but not the least I thank my parents and my friend for their moral support and valuable help.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	3
	LIST OF TABLES	9
	LIST OF FIGURES	10
	LIST OF ABBREVIATION	12
1	**INTRODUCTION**	13
2	**LITERATURE SURVEY**	15
3	**DATASET DESCRIPTION**	19
	3.1. IDENTIFIERS AND METADATA	19
	3.2. RATINGS AND VOTES	20
	3.3. TEXTUAL DATA	20

	3.4. VERIFICATION AND MODERATION	20
	3.5. BEHAVIORAL AND TEMPORAL FEATURES	21
4	**SYSTEM DESIGN AND ARCHITECTURE**	22
	4.1. DATA PRE-PROCESSING AND COLUMN OPTIMIZATION	22
	4.2. MULTI-LAYER VERIFICATION AND FILTERING	22
	4.3. CONTENT QUALITY ASSESSMENT	23
	4.4. FAKE REVIEW DETECTION MODEL USING BERT-LSTM-CNN	23
	4.5. RATING CORRELATION	24
	4.6. HELPFULNESS SCORE OPTIMIZATION	24
	4.7. CHATBOT DEVELOPMENT	25

	4.8. ARCHITECTURE DIAGRAM	26
	4.9. FLOW DIAGRAM	27
5	**MODEL TRAINING**	28
	5.1. FEATURE SELECTION	28
	5.2. TEXT FEATURE TRANSFORMATION USING BERT EMBEDDINGS	28
	5.3. NUMERICAL FEATURE PROCESSING	29
	5.4. FEATURE COMBINATION	29
6	**EXPLAINABILITY**	30
	6.1. FEATURE CONTRIBUTIONS TO MODEL DECISIONS	30
	6.2. HOW THE MODEL DETECTS FAKE REVIEWS	32

	6.3. MODEL INTERPRETABILITY: FEATURE IMPORTANCE IN DECISION-MAKING	32
	6.4. DECISION-MAKING FOR SPECIFIC SCENARIOS	32
	6.5. EXPLAINABILITY FOR PREDICTIONS	34
7	**EXPERIMENT AND ANALYSIS**	36
8	**EVALUATION**	38
9	**SOCIAL IMPACT**	44
10	**CONCLUSION**	46
11	**FUTURE WORK**	48
12	**REFERENCES**	50

LIST OF TABLES

TABLE NO	DESCRIPTION	PAGE NO
Table 8.1	Evaluation table for different LLM'S and SLM's.	39
Table 8.2	Showing Benchmark for AI Chatbots.	40

LIST OF FIGURES

FIGURE NO	DESCRIPTION	PAGE NO
Figure 4.1	Showing Architecture Diagram.	26
Figure 4.2	Showing Flow Diagram.	27
Figure 7.1	Showing Application UI with an example review.	36
Figure 7.2	Showing predicted fake review with explainability.	36
Figure 7.3	Showing predicted genuine review with explainability.	36
Figure 7.4	Showing TrustScore chatbot interface.	37
Figure 7.5	Showing the dataset after classification.	37
Figure 8.1	Showing Training and Validation Accuracy for BERT-LSTM-CNN.	38
Figure 8.2	Showing Training and Validation Loss for BERT-LSTM-CNN.	38
Figure 8.3	Showing Effect of Helpness Score.	38

Figure 8.4	Showing model quality of different language models.	38
Figure 8.5	Bar Chart showing Ease of Use of TrustScore.	40
Figure 8.6	Bar Chart showing Response Satisfaction of TrustScore.	41
Figure 8.7	Pie Chart showing Error Rate of TrustScore.	41
Figure 8.8	Pie Chart showing Response Speed of TrustScore.	41
Figure 8.9	Bar Chart showing Response Accuracy of TrustScore.	42
Figure 8.10	Pie Chart showing Task Completion Rate of TrustScore.	42
Figure 8.11	Bar Chart showing Speed and Efficiency of TrustScore.	42
Figure 8.12	Bar Chart showing Retention Rate of TrustScore.	43
Figure 8.13	Pie Chart showing performance comparison of TrustScore.	43

LIST OF ABBREVIATIONS

NO.	ACRONYM	ABBREVIATIONS
1	KNN	K-Nearest Neighbours
2	SVM	Support Vector Machines
3	NLP	Natural Language Processing
4	LSTM	Long Short-Term Memory
5	CNN	Convolutional Neural Network
6	RNN	Recurrent Neural Network
7	BERT	Bidirectional Encoder Representations from Transformers
8	LLM	Large Language Model
9	SLM	Small Language Model
10	RAG	Retrieval-Augmented Generation

1. INTRODUCTION

Online comments play an important role in influencing what one purchases. The increasing number of fake and manipulative reviews on e-commerce websites, however, has raised concerns regarding transparency and credibility in online marketplaces. Based on the need to ensure the authenticity and functionality of product reviews on e-commerce websites, this paper introduces a multi-layered fake review detection system based on machine learning and AI-powered chatbot conversation [1], [2].

To ensure maximum input data for training models, the verification process begins with systematic data preprocessing, encompassing format normalization, text sanitization, and dataset balancing. To enhance review credibility, a multi-layered filtering system is employed. This involves IP tracking, which prevents several reviews of the same item from being sourced from the same location, cutting down on spam and fraud, and purchase verification, which ensures that only reviews associated with verified purchases are retained [6].

It deploys a combined deep learning algorithm that integrates CNN, LSTM, and BERT to classify the reviews as authentic or fake with the aim of enhancing reliability even further. The CNN detects phrases' patterns with high frequency rates in fake reviews, LSTM preserves long-term dependence, and BERT captures context embeddings. Class imbalance is alleviated by employing Focal Loss to optimize the model,

and this ensures fraud reviews are adequately identified [7].

An index of helpfulness score based on user interaction metrics is also employed to further sort reviews, prioritizing those that are helpful and informative. Customers are assured of being able to access the most trustworthy product reviews due to this adaptive ranking system [8].

Besides classification, the system has a chatbot based on the Phi-3 modified model that assists users in searching for relevant product information. The chatbot is able to fetch and analyze authenticated product reviews since it is built using Retrieval-Augmented Generation (RAG) and ChromaDB, a vector database. Users no longer need to sort through reviews manually because it provides fact-based responses to their inquiries regarding the usability, durability, and performance of products [9].

By means of user feedback cycles and continuous learning mechanisms, the system continually adapts to evolving fraudulent strategies, guaranteeing long-term effectiveness in detecting and diminishing fraudulent reviews. AI-driven chatbot assistance and this multi-aspect verification process encourage greater trust, transparency, and well-educated e-commerce decision-making, to the advantage of customers and ethical sellers. The ultimate impact is a more trustworthy online marketplace in which genuine customer feedback informs purchasing decisions and enhances the overall efficacy and integrity of online commerce.

2. LITERATURE SURVEY

Recent years have seen considerable importance to detect fake reviews given its critical role in ensuring the reliance of web applications. This problem was approached by researching machine learning methods successfully in a number of studies.

Punde et al. (2021) proposed a machine learning-based method to classify reviews as genuine or deceptive, focusing on the sentiment of the review. Results from the study showed that deep learning models are more effective compared to traditional methods, and emotional tone is an important attribute in determining misleading feedback. The authors utilized an actual Yelp dataset to test and validate their outcomes, making use of classifiers such as K-Nearest Neighbours (KNN), Support Vector Machines (SVM), and the like [1].

Techniques in the domain of machine learning have in fact been exploited by Mohawesh et al. in 2021 to conduct a review on the detection of false reviews. Here, the best performance has been achieved while n-gram feature extraction techniques were paired with classifiers like logistic regression and SVM [2]. A couple of drawbacks, which surfaced in the course of the study, were the inclusion of subtle aspects of reviewer behavior to improve precision in detection.

Gayathri et al. (2023) offered a new method that analyzed the behavioral characteristics of the reviewer, comprising the length of review and frequency of submission for fraudulent reviews. From their

work, SVM classified reviews better than classifiers that demonstrated how the inclusion of behavioral data adds significance to enhance model performance [3].

In a different exploration, Gayathri et al. (2023) used the Yelp dataset to look deeper into machine learning techniques applied for detection and removal of fake reviews. The preprocessing techniques applied involved tokenization and elimination of stop words, followed by TF-IDF techniques to extract features in this study. The results were presented in how well these XGBoost and SVM classifiers do work in identifying spam comments with great feature selection information meant for higher accuracy detection [4].

Jindal and Liu (2018) developed a novel method of fraudulent review detection by exploiting behavioral patterns and linguistic features [5]. Their results showed that incorporating user activity patterns along with content-based features increased detection accuracy significantly, resulting in 85% success in detecting fabricated reviews.

A novel technique to identify fraudulent review patterns by applying time-series analysis was developed by Kumar et al. (2020) [6]. Suspicious review patterns, according to their analysis, often emerged at specific times in certain temporal clusters, especially at the time of rival debuts or during promotional days. The technique obtained a 92% accuracy rate in the identification of coordinated phoney review campaigns by using sophisticated statistical techniques to detect

aberrant review bursts.

Wang and Chen (2022) presented an aspect-based sentiment analysis approach to measure the semantic consistency of reviews. The measure presented estimated the correlation between expressing sentiments and product attributes, and so the authors identified that phoney reviews generally involved the generic or mismatched pairings of sentiment-feature. The semantic approach portrayed that the ability to detect fraudulent reviews with 88% accuracy could be improved especially in situations when traditional approaches fall short of circumstances [7].

Rodriguez et al. (2023) employed GCNs with semantic similarity metrics for fraudulent reviews detection. Their approach was 7% more accurate than the traditional methods, by a huge margin. The analysis proved to be highly effective in detecting subtle phoney reviews that maintained linguistic authenticity but conveyed dubious trends in user behavior networks [8].

Singh and Patel (2023) proposed a hybrid approach that incorporates the analysis of reviewer behavior with NLP [9]. The proposed system monitored factors such as emotional consistency, semantic patterns, and review density. A comparison with single-feature techniques and varied characteristics ensures an increase in detection accuracy by 15%.

Thompson et al. (2024) introduced the idea of a distinction between

human-written and AI-generate reviews using transformer models [10]. Interestingly, in their study, by introducing a training phase, they proved that it is possible to overcome the problem related to robot-created reviews with advancement in language models, reaching a correct recognition level of about 94% for machine-generated fake reviews.

Advanced deep-fake review identification algorithms, combined with natural language processing capabilities, provide an effective interactive solution for consumers to access reliable information about a product. The hybrid solution enables customers to pose queries to receive accurate responses regarding their inquires into goods and services, all the while offering conversational AI and real-time review authenticity analysis as an interface with customers.

3. DATASET DESCRIPTION

The data isn't small-time—it consists of 88,551 records with 17 columns full of information regarding customer feedback. The data is centered on products from the Mobile and Electronics category, and it combines textual and numerical data. It has everything from user sentiments and ratings to user behavior patterns.

Key Features in the Dataset

3.1. Identifiers and Metadata

- **Marketplace**: Every review in this dataset comes from the *US* marketplace, representing the geographic location where the reviews were posted.
- **Unique Identifiers**:
  customer_id, review_id, product_id, product_parent: These act like tags to trace specific customers, reviews, products, and product groups. They're super useful for tracking individual reviews and grouping related products for deeper analysis.
- **Product Info**:
  - **product_title**: The name of the product being reviewed.
  - **product_category**: All entries fall under one category—*Mobile_Electronics*.

3.2. Ratings and Votes

- **Star Rating**:
  - Each review includes a customer's rating on a 1-to-5-star scale.
  - On average, the ratings lean positive, with a mean of 3.75, and most reviews land around 4 or 5 stars.
- **Helpfulness Metrics**:
  - **helpful_votes**: Shows how many people found a review helpful.
  - **total_votes**: Reflects the total number of votes a review received.
  - On average, reviews have 1.28 helpful votes, but there are outliers—some reviews scored up to 769 helpful votes! These features play a big role in identifying which reviews are more reliable.

3.3. Textual Data

- **Review Headline**: A short, attention-grabbing summary for the review (e.g., "Very Happy!" or "Five Stars"). The dataset includes tons of unique headlines, showing a lot of variety.
- **Review Body**: The detailed part of the review. With 86,126 unique entries, this is where most of the action happens for textual analysis. This variability gives the model plenty of material to work with when looking for patterns.

3.4. Verification and Moderation

- **Vine Program**: Some reviews are part of the Vine program, which means they may be incentivized (marked as "Y" for yes, "N" for

no).

- **Verified Purchase**: Out of the entire dataset, 41,296 reviews are tied to verified purchases. These reviews are generally more trustworthy since they show the customer actually bought the product.

3.5. Behavioral and Temporal Features

- **IP Address**: Tracks where the review came from. This is key for spotting spam, such as duplicate reviews from the same source.
- **Review Date and Timestamp**: These provide a timeline for when each review was posted. They're perfect for analyzing trends or detecting coordinated activity (like a sudden flood of reviews).

This mix of ratings, metadata, text, and behavioral patterns gives the model a comprehensive toolkit for detecting fake reviews and analyzing genuine ones with confidence.

4. SYSTEM DESIGN AND ARCHITECTURE

The proposed system is built in two phases: an AI-powered chatbot for user interface and machine learning-based detection of fake reviews. The Phi-3 modified chatbot is implemented to assist users in querying for review-related information, and the BERT-based deep learning model is employed to detect reviews as real or fake. Data preprocessing, training of the model, review classification, and integration of the chatbot are the different phases of the methodology.

4.1 Data Pre-processing and Column Optimization

The data is a large set of reviews, both genuine and fake, with the proper labels. Text processing methods are applied to prepare the data for training. To ensure a uniform input length, this involves removing special characters, making the text lowercase, tokenisation, and padding. The BERT tokeniser is applied to tokenise the text data, converting every review into a sequence of tokens that can be fed into the model. In addition, to ensure that both fake and real reviews are equally represented throughout the training process, dataset balancing techniques such as undersampling and oversampling are utilized [5].

4.2 Multi-Layer Verification and Filtering

Before it is classified, the system employs multi-layer verification filtering to enhance review credibility. The initial filter eliminates unverified submissions by ascertaining if a review has been linked to a

verified purchase. To identify and mark multiple reviews from one source, IP address tracking is then implemented. Additionally, category matching ensures that a review has been linked to the correct product category. By doing so, the risk of spam and fake reviews is highly minimized, ensuring genuine feedback for analysis [8].

4.3 Content Quality Assessment

Sophisticated screening techniques are utilized in content quality control to analyze various quality markers in reviews. Profanity filter algorithms are used by the system to remove insulting words and spam detection tools to remove machine-produced or mass posts. Text quality evaluation ensures criticism is insightful and constructive by reviewing if a review is reasonable and pertinent to specifications of a product [8].

4.4 Fake Review Detection Model using BERT-LSTM-CNN

The model employs a hybrid deep learning architecture consisting of CNN, LSTM, and BERT to classify reviews as real or fake. LSTM is employed to learn sequential relationships once rich contextual embeddings have been learned from the review text using the BERT model. A CNN layer then further processes the output by detecting phrase-level patterns that are commonly found in fake reviews. A softmax activation function is implemented in a fully connected layer for the final classification, where it predicts whether a review is real (0) or fake (1) [6].

During training, the model is minimized with the Adam optimizer, and Focal Loss is used to combat class imbalance by placing greater emphasis on hard-to-classify counterfeit reviews.The model is subjected to multiple training epochs, and accuracy, loss, and confusion matrix evaluation are employed to measure performance. Post-training, the model is saved and deployed for real-time review classification [7].

4.5 Rating Correlation

Text sentiment extraction analyzes the emotional content of review material, while rating correlation methods are only comparable to sentiment scores and star ratings. By calling out reviews in which the sentiment of the text and the numerical ratings do not coincide, the method ensures stringent authenticity and reliability [9].

4.6 Helpfulness Score Optimization

A helpfulness score (HS) is calculated using user engagement metrics to rank insightful reviews. The following formula determines the score:

$$\text{HS} = \frac{Number\ of\ Helpful}{Total\ Votes}$$

Higher helpfulness-rated reviews appear first, which provides users quick access to dependable and useful comments. The feature of the system to identify most informative reviews is constantly enhanced through dynamic ranking updating based on user interactions in real time [10].

4.7 Chatbot Development

A Phi-3 modified model-driven chatbot is implemented to assist users in fetching relevant product information after the ML model for detecting fake reviews has been implemented. The chatbot relies on ChromaDB, a vector database, for storing and fetching authenticated reviews. It was implemented using Retrieval-Augmented Generation (RAG). It can be effortlessly implemented in a front-end app, ensuring an intuitive and user-friendly interface [10].

The chatbot is designed to return accurate, fact-supported answers to user queries on product reviews. The chatbot seeks relevant verified reviews stored in ChromaDB whenever a user inquires about the usability, longevity, or functionality of a product. The Phi-3 improved model is thereafter utilized to reason over the information that was accessed, generating an answer that is contextually sensitive and logical. Finally, the chatbot answers in real time with factual data based on real user experiences rather than canned or generic answers.

One of this chatbot's greatest strengths is the capacity to rapidly locate relevant reviews, ensuring users obtain the best possible product information. It gives real-time responses, responding to questions and delivering data in milliseconds. By tailoring answers to the specific needs of the user, context-aware analysis enhances decision-making and ensures insights provided match real product experiences. In addition, since the chatbot is fully embedded in a frontend application,

users are able to communicate with it naturally through an entertaining and captivating interface.

This chatbot ensures that customers can make well-informed buying decisions by leveraging retrieval mechanisms powered by AI and authentic user experiences. It is a reliable virtual assistant, rescuing customers from the need to sift through many reviews manually and providing them with quick, relevant, and insightful responses based on real product facts.

4.8. ARCHITECTURE DIAGRAM

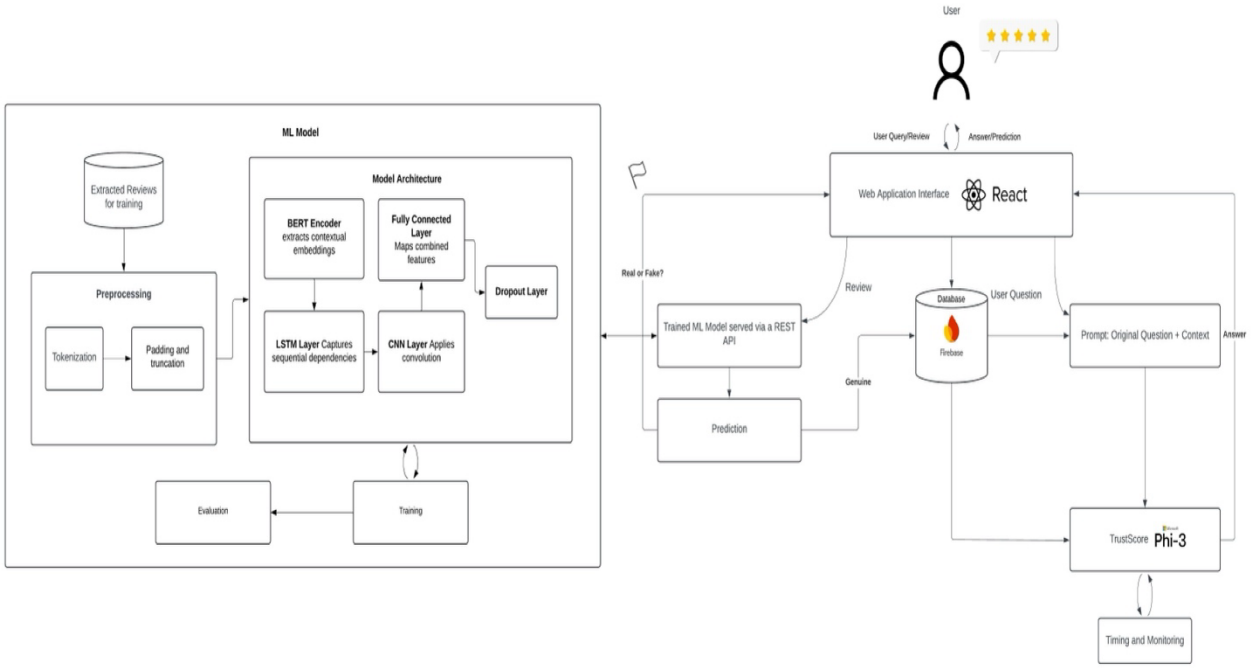


Figure 4.1

4.9. FLOW DIAGRAM

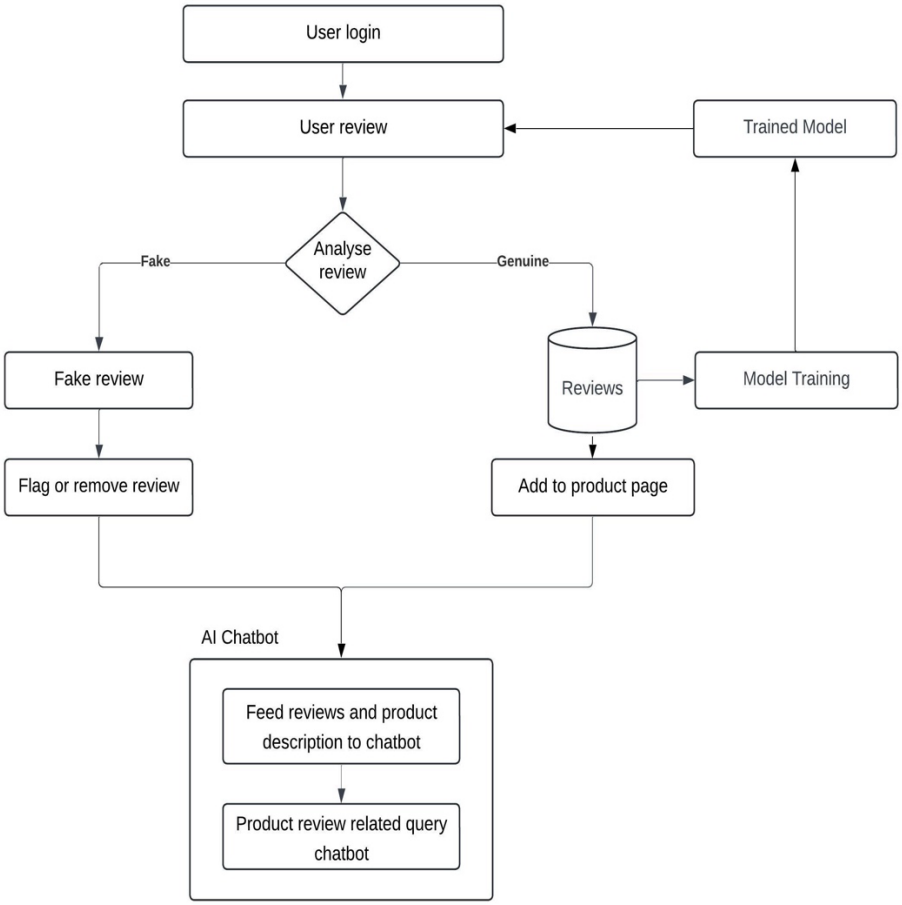


Figure 4.2

5. MODEL TRAINING

By using a mix of deep learning and feature engineering methods, the training phase of the model is meant to effectively distinguish between imitations and original assessments. The BERT-LSTM-CNN model is trained with a preprocessed dataset that consists of well-selected and converted textual and numerical features to enhance the precision of classification.

5.1 Feature Selection:

A feature engineering process is employed to remove redundant and useless properties once the dataset is separated into features (X) and target variable (Y), where Y indicates whether a review is real or not. Removing features such as sentiment_match, failed_constraints, duplicate_review, and helpfulness_score avoids storing useless attributes in model training and avoids data leakage.

5.2 Text Feature Transformation using BERT Embeddings:

The method utilizes BERT embeddings to transform review text into numeric forms instead of the traditional TF-IDF vectorisation. By extracting strong semantic meaning from transformer-based contextual embeddings, BERT improves the ability of the model to tell real and fake evaluations apart. For sequence analysis and pattern recognition, pre-trained BERT tokenizes review text, maps it into dense vector

representations, and then inputs these embeddings to the CNN and LSTM layers.

5.3 Numerical Feature Processing:

The model utilizes numeric structured features derived from non-text attributes as well as textual embeddings. Columns that are not useful to determine review validity, such as customer_id and product_id, are removed. StandardScaler is utilized to standardize relevant numeric features (e.g., helpfulness_score, review_length, and verified_purchase), ensuring consistent distributions and better machine learning model compatibility.

5.4 Feature Combination:

BERT-text embedding (dense matrix) and scaled numeric features (sparse matrix) are concatenated using hstack (horizontally stack) to create a unified feature set. This union allows the model to leverage both structured information and text context to enhance its performance in differentiating between fake and authentic reviews.

The multi-layered validation technique for reviews is based on the model's robust generalisation and high reliability in detecting fake reviews, which are attained by fusing deep learning (BERT-LSTM-CNN) with structured numerical analysis.

6. EXPLAINABILITY

It is important to understand the decision-making process of a machine learning model, especially when it involves identifying fake reviews. This section breaks down the decision-making process of the BERT-LSTM-CNN model, explaining the key features, detection processes, and interpretability of the model that result in accurate predictions.

6.1. Feature Contributions to Model Decisions

The model leverages both numerical and textual features, each playing a unique role in determining whether a review is genuine or fake:

1. **Verified Purchase:**

Reviews linked to a verified purchase are more likely to be genuine. Non-verified reviews often exhibit spam-like patterns and are more susceptible to manipulation, making this a critical feature in classification.

2. **BERT Embeddings for Text Analysis:**

The system replaces traditional TF-IDF with BERT embeddings, allowing the model to capture deeper contextual meaning in reviews. Suspicious patterns, such as repeated promotional phrases (*"best deal," "must buy"*) or sentiment-ratings mismatch, are detected more effectively using deep language understanding.

3. **Sentiment Alignment:**

The system compares the textual sentiment of a review with its star rating. For example, a 5-star review that contains negative words like "terrible" or "worst" is flagged as inconsistent, contributing to fraud detection.

4. **Helpfulness Score:**

The model evaluates the helpfulness score, calculated as the ratio of helpful votes to total votes. Reviews with higher helpfulness scores are considered more credible, while reviews with low or no engagement may be less reliable.

5. **IP Address Behavior:**

If multiple reviews originate from the same IP address within a short timeframe, it raises red flags indicating review manipulation or bot-generated content.

6. **Purchase Relevance:**

The model ensures that reviews match the product category. For instance, a review mentioning clothing for an electronics item is marked as irrelevant, improving content validity.

6.2. How the Model Detects Fake Reviews

1. Dual Sentiment Detection:

Using BERT-based sentiment analysis, the model flags inconsistencies within a review's text and its overall sentiment trend. A review containing both extreme praise and criticism is marked as suspicious.

2. Behavioral Patterns:

The model tracks patterns of repeated reviews from the same customer or IP address. If multiple near-identical reviews are posted for the same product, it signals fraudulent activity.

3. Spam and Profanity Filtering:

The system scans review text for spammy content, promotional phrases, and offensive language. Reviews with excessive repetition of promotional keywords (*"limited offer," "best price"*) or inappropriate words are flagged as potential spam.

6.3. Model Interpretability: Feature Importance in Decision-Making

The system assigns weights to features, helping interpret how each attribute impacts the classification of a review. Key contributing features include:

- **Verified Purchase** (High weight – critical indicator of authenticity)

- **Sentiment Alignment** (Identifies mismatches between text and ratings)
- **Helpfulness Score** (Determines review credibility based on user engagement)
- **BERT Embeddings** (Deep text analysis for detecting manipulation)
- **IP Address Behavior** (Flags suspicious activity)
- **Star Rating Consistency** (Ensures coherence between numerical ratings and review text)

Less critical elements, such as product categories, hold lower priority in classification.

6.4. Decision-Making for Specific Scenarios

Case 1: Genuine Review

A review is confidently classified as authentic if it meets the following criteria:

- Verified purchase confirmed
- Sentiment matches star rating
- High helpfulness score
- No spam or repeated patterns detected

Case 2: Fake Review

A review is flagged as fake if:

- The sentiment contradicts the star rating (e.g., "Worst product ever" with 5 stars).
- Multiple identical reviews originate from the same IP address.
- It contains excessive promotional terms or offensive language.

6.5. Explainability for Predictions

To ensure transparency in decision-making, the model logs and analyzes the following components for each review:

1. **Feature Values:**

   Example: A verified purchase is marked as 1, and a helpfulness score of 0.8 is assigned, contributing to classification.

2. **Intermediate Calculations:**
   - Sentiment scores from textual analysis.
   - Behavioral checks for IP duplication or spam detection.

3. **Feature Weights:**

   The relative importance of each feature in the final decision.

4. **Final Decision:**

   Whether the review is classified as genuine or fake, along with a confidence score indicating prediction certainty.

By integrating BERT-LSTM-CNN for deep sentiment analysis and multi-layered verification mechanisms, the system enhances review authenticity detection while ensuring explainability and trustworthiness in its predictions.

7. EXPERIMENTS AND RESULTS

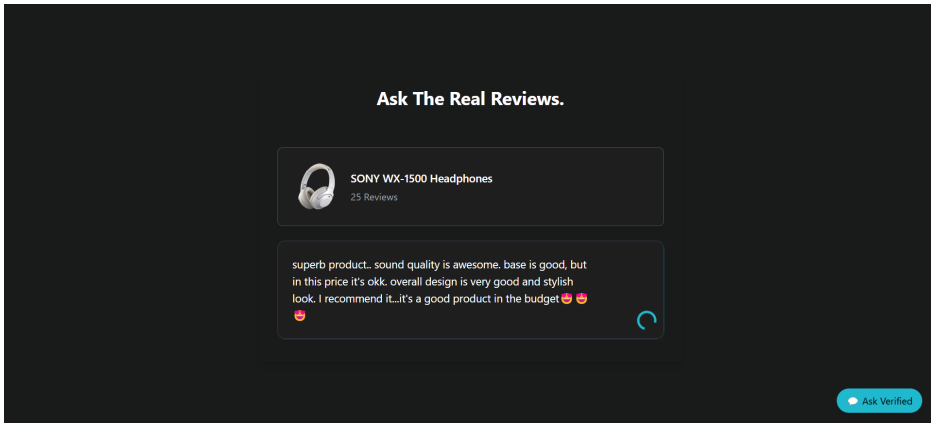


Figure 7.1

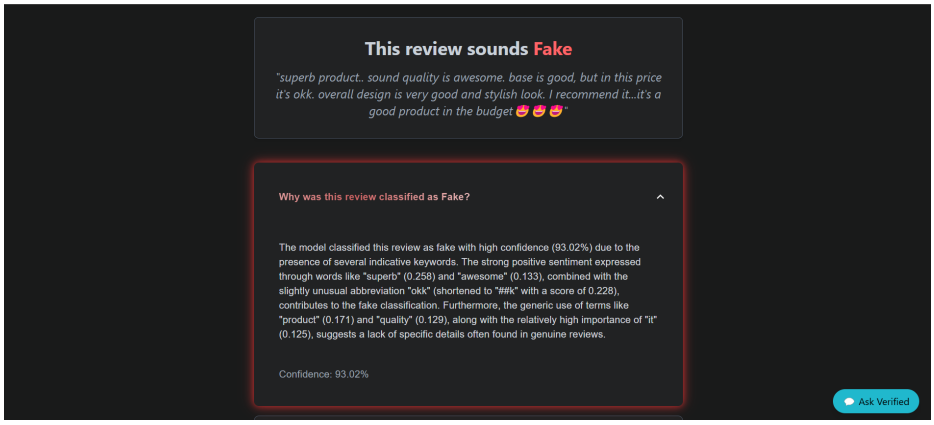


Figure 7.2

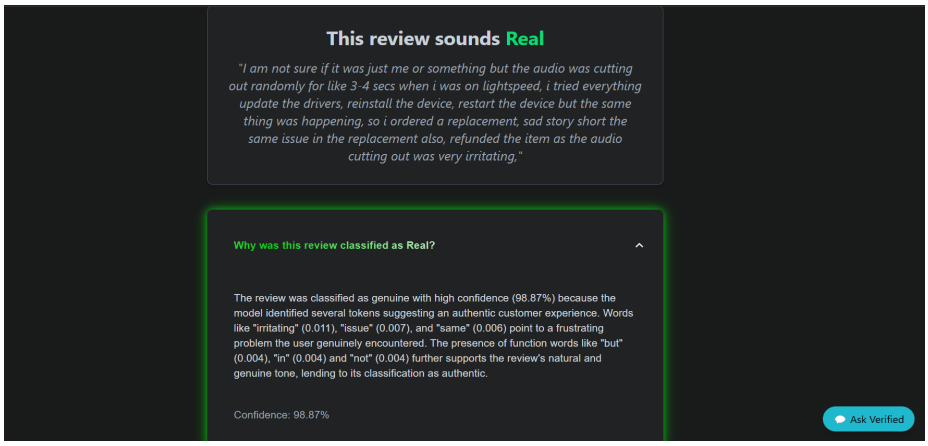


Figure 7.3

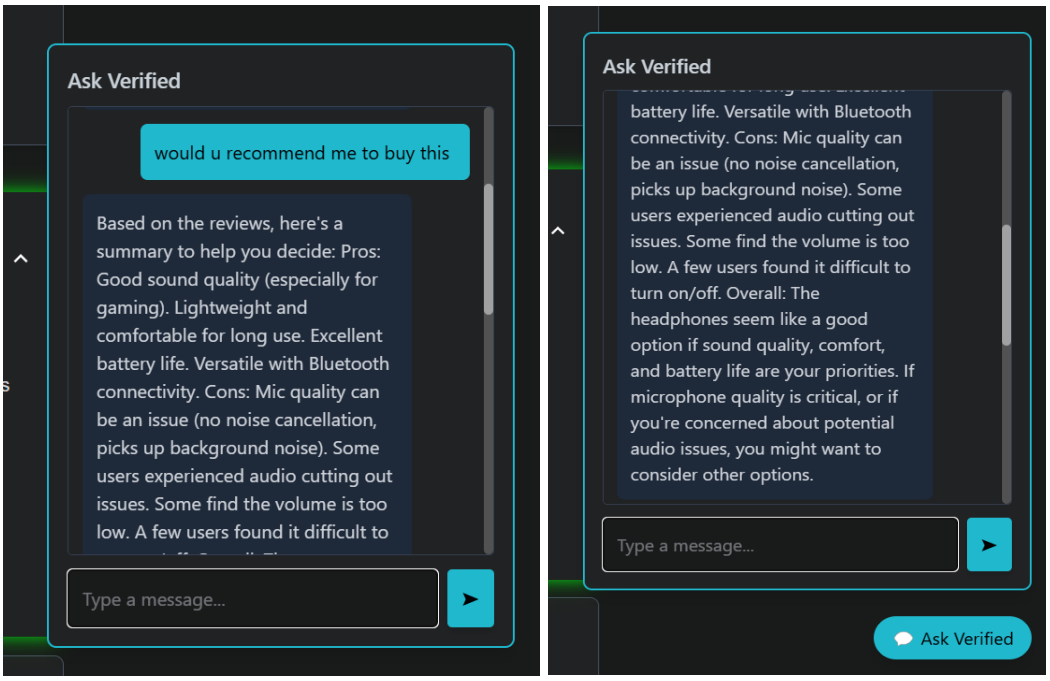


Figure 7.4

```
Total number of rows: 88578
Number of rows where 'genuine' is 0: 47267
Number of rows where 'genuine' is 1: 41296
```

genuine	failed_constraints	duplicate_review	sentiment_match	helpfulness_score
1	[]	0	1	0.990099
1	[]	0	1	0.988235
1	[]	0	1	0.984848
1	[]	0	1	0.977778
1	[]	0	1	0.977778

Figure 7.5

8. EVALUATION

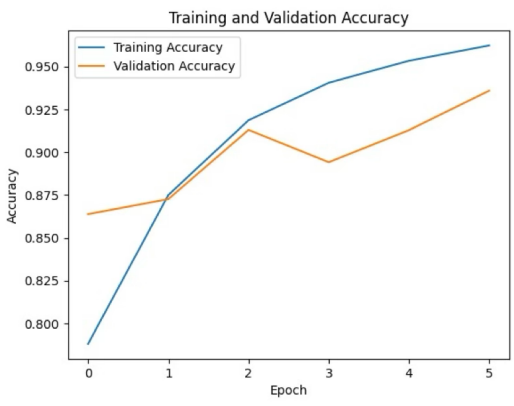


Figure 8.1

Figure 8.2

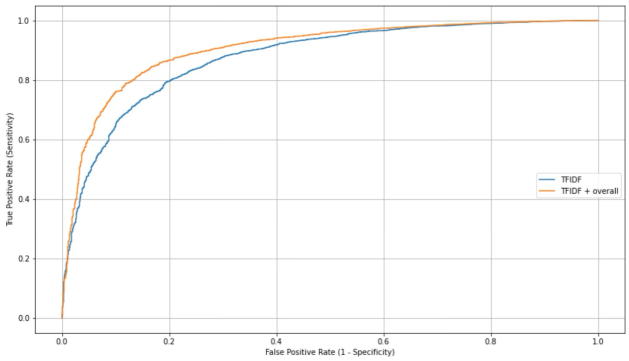
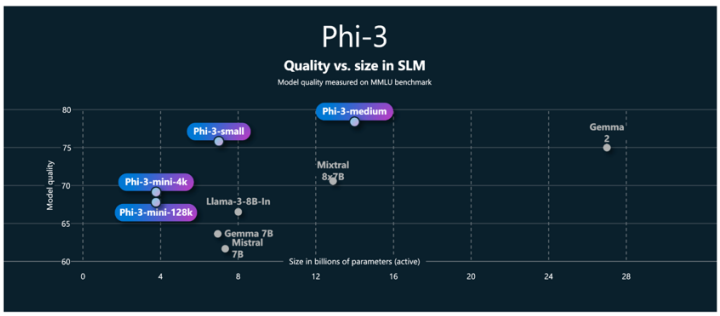


Figure 8.3



Phi-3 Mini

Phi-3 Mini is a 3.8B parameters, lightweight, state-of-the-art open model trained with the Phi-3 datasets that includes both synthetic data and the filtered publicly available websites data with a focus on high-quality and reasoning dense properties.

Figure 8.4

Model	Local Runtime Speed	Model Size & Efficiency	Inference Performance	Memory Requirements	Optimized for Small-Scale AI?	Suitability for Review-Based Chatbot
Phi-3 Mini (4B)	Very fast (low latency, optimized for local deployment)	Small, lightweight (4B parameters, designed for efficiency)	Fast inference, optimized for low-resource environments	Lower VRAM requirement (~6GB for 4-bit quantization)	Yes, designed for lightweight AI applications	Excellent – fast, context-aware, and ideal for review analysis
Mistral 7B	Slower than Phi-3 Mini (optimized for broader general tasks)	Efficient, but larger than Phi-3 Mini	Strong inference but requires more VRAM	Higher VRAM (~12GB for smooth operation)	Suitable for general AI applications	Good, but slower for localized chatbot tasks
LLaMA 3 (8B/70B)	Slower than both Phi-3 Mini and Mistral	Large models (8B-70B parameters), requiring high-end GPUs	High accuracy but slow for local inference	Very high VRAM usage (~16GB for 8B, 48GB+ for 70B)	No, better for cloud-based deployments	Too heavy for local chatbot processing
DeepSeek 7B	Slower local inference compared to Phi-3 Mini	Efficient, but still requires more VRAM than Phi-3 Mini	Strong performance but not as optimized for lightweight AI	Needs ~12GB VRAM for smooth local execution	Designed for multilingual and general-purpose AI	Decent, but requires higher computational power

Table 8.1

Why Phi-3 Mini (4B) is the Best Choice for This Project:

- **Fastest Local Runtime Speed** – Phi-3 Mini is optimized for low-latency, real-time execution on local machines, making it ideal for chatbot applications.
- **Low Memory Requirements** – Unlike LLaMA 3 or Mistral, Phi-3 Mini can run smoothly on mid-range GPUs (~6GB VRAM) or even high-end CPUs, ensuring efficiency.

- **Optimized for Small-Scale AI** – Phi-3 Mini is designed for local AI applications, unlike LLaMA 3 (optimized for cloud) or DeepSeek (multilingual-heavy, requiring more resources).
- **Best for Review-Based Chatbots** – Phi-3 Mini's retrieval-augmented generation (RAG) and fast token processing make it highly effective for answering user queries based on verified reviews, improving accuracy and responsiveness.

Benchmark for a AI Chatbot:

Metric	Description	Benchmark/Good Score
User Satisfaction	Average score from satisfaction questions.	4/5 or above
Response Accuracy	Percentage of queries answered correctly.	85% or above
Response Time	Average time taken to respond.	< 3 seconds
Task Completion Rate	Percentage of users who achieved their goal.	80% or above
Error Rate	Number of failed or misunderstood queries.	< 10%
Retention Rate	Percentage of users willing to return.	70% or above

Table 8.2

SURVEY RESULTS:

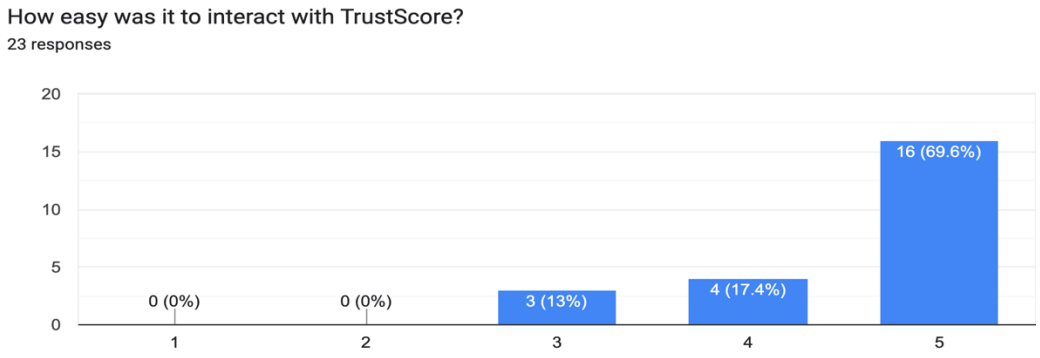


Figure 8.5

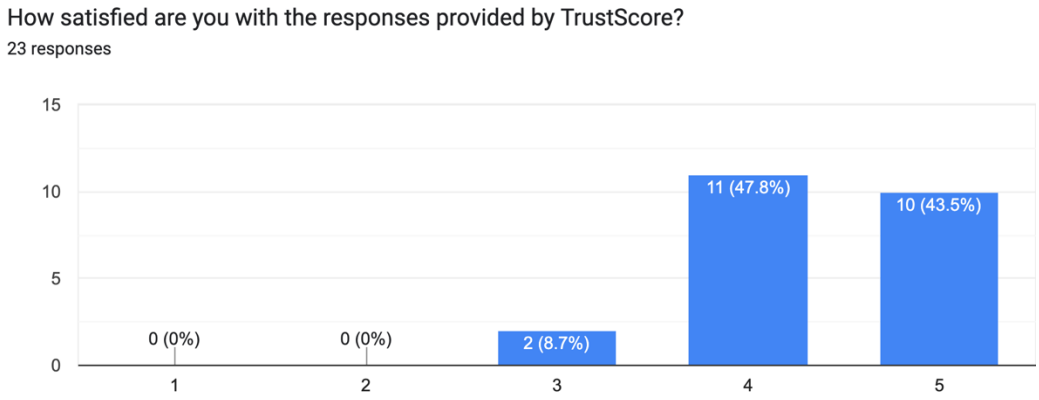


Figure 8.6

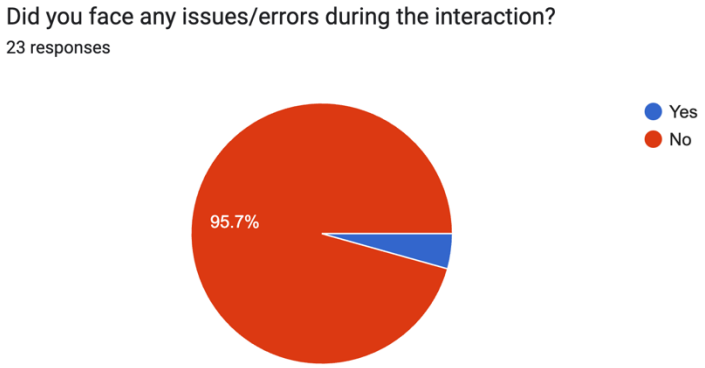


Figure 8.7

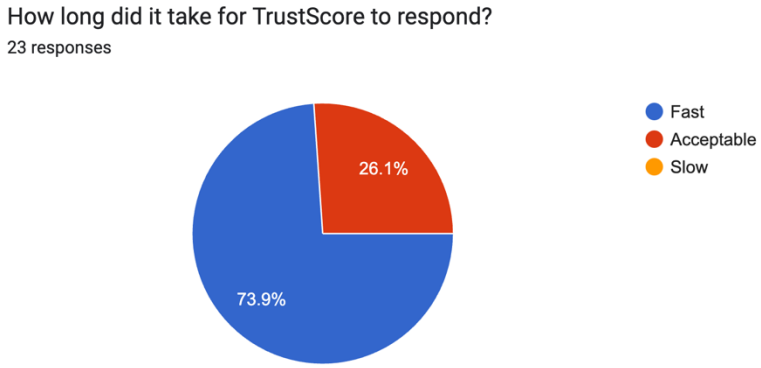


Figure 8.8

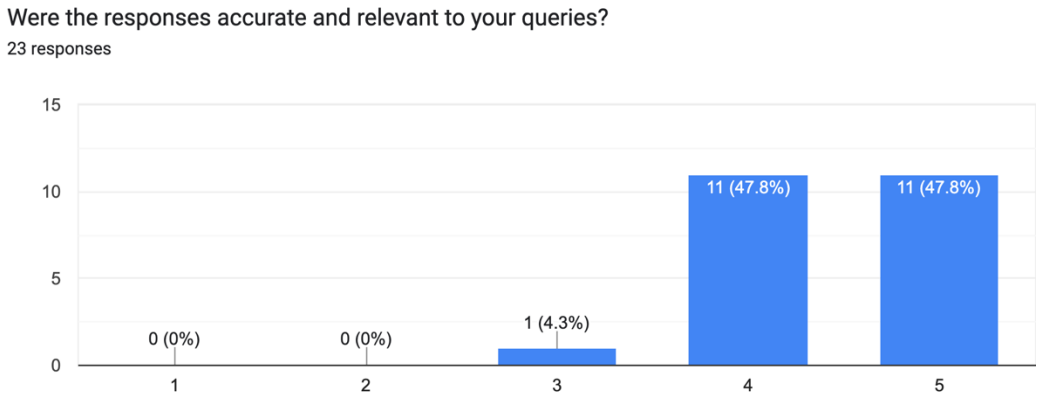


Figure 8.9

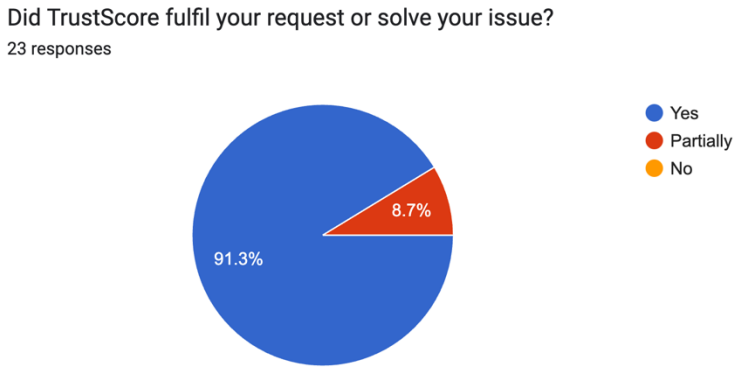


Figure 8.10

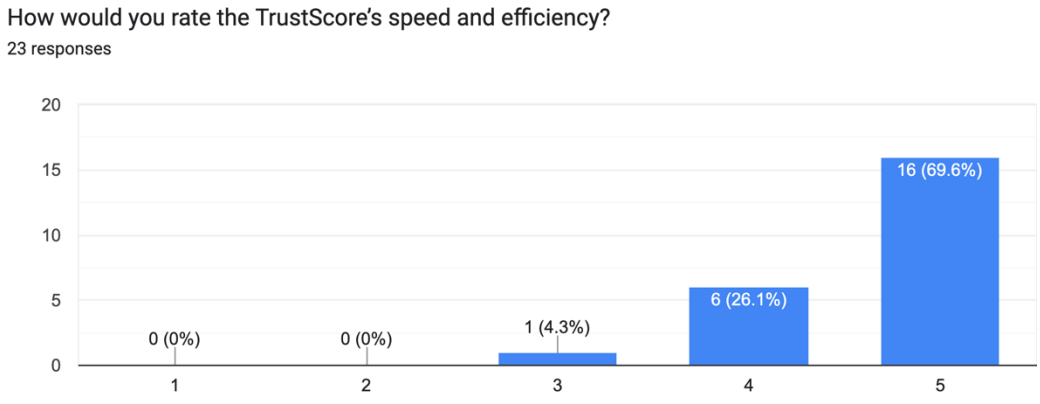


Figure 8.11

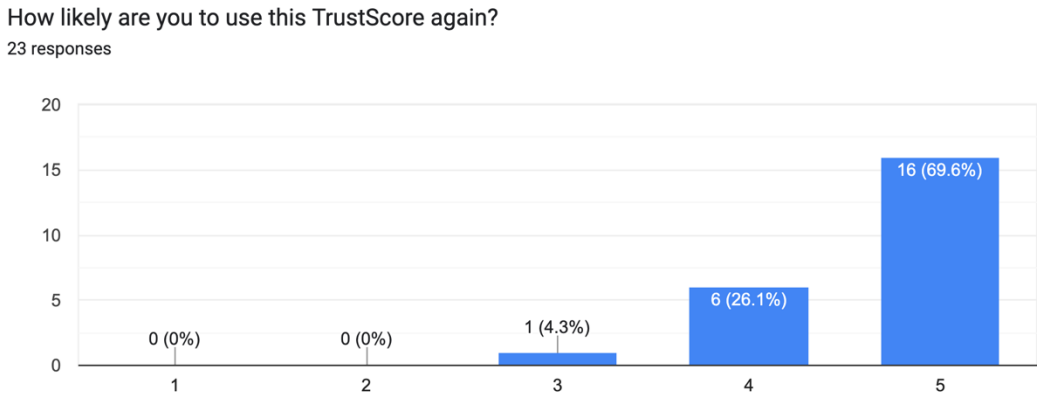


Figure 8.12

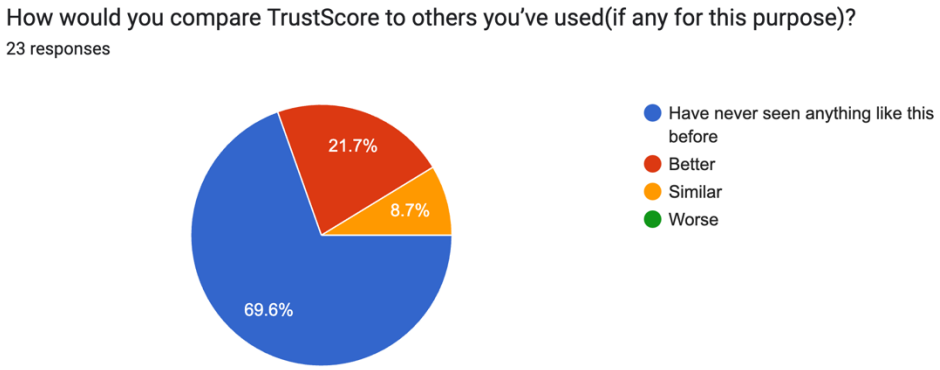


Figure 8.13

9. SOCIAL IMPACT

The proposed approach redefines trust and transparency in online commerce and carries significant social and business implications. This approach enhances customer trust by creating a more credible review environment, which promotes honest business-to-consumer interactions. Eliminating fake ratings and stimulating genuine user feedback enhances marketplace integrity online and renders the virtual economy more truthful and accountable.

By ensuring customers get authentic and fact-based product information, the system enables more informed decision-making beyond a single transaction. Consumers and sellers both benefit from higher customer satisfaction, lower return rates, and improved consumption patterns. Ultimately, a more efficient and sustainable marketplace is achieved when consumers make informed buying decisions and companies lose less due to false assessments.

The system has impact in emerging regions too, where it can facilitate ethical e-commerce and online literacy. It discourages manipulative marketing practices and encourages ethical customer conduct by setting a higher standard for review authentication. This project could shape how e-commerce websites authenticate and showcase user-generated content in the future and serve as an example of digital trust protocols.

The technology is also a defence barrier for customers and businesses. It shields businesses from unethical competition, dishonest review farming, and deceptive rating inflation by providing them with access to verified product experiences and AI-based insights. Only authentic reviews drive buying decisions due to the integration of multi-layered verification, chatbot-powered review retrieval, and AI-based sentiment analysis.

All things being equal, this approach provides a new benchmark for digital trust and review legitimacy, paving the way for an online economy that's more open, efficient, and ethical. It transforms the online shopping experience by leveraging AI, machine learning, and formalized verification processes, which is beneficial to businesses, individual consumers, and the overall e-commerce ecosystem.

10. CONCLUSION

As far as ensuring the accuracy and reliability of online product reviews, our review validation and chatbot integration approach is a breakthrough. Through a multi-step filtering process that encompasses data preprocessing, purchase authentication, IP validation, and category relevance checks, the system systematically excludes content that is not credible. In addition, by ensuring that only genuine and insightful reviews are retained, spam filtering and profanity detection maintain the highest levels of professionalism, which works to the advantage of both buyers and sellers.

Apart from simple filtering, the system employs BERT for advanced sentiment analysis, complemented by the semantic abilities of WordNet. Cross-referencing star ratings with sentiment scores by this process ensures higher authenticity and enables thorough contextual comprehension of review content. The system boosts the authenticity of the review corpus by eliminating low-quality or deceptive reviews by prioritizing content quality.

One of the key benefits of this system is its ingenious chatbot integration. With the assistance of Retrieval-Augmented Generation (RAG) and ChromaDB, the Phi-3 enhanced chatbot leverages the verified review database to supply users with context-specific, fact-based answers. The chatbot is accurate and insightful in its response through prioritizing high helpfulness reviews, thus retaining real

customer experiences. Customers feel more assured because of this, and they are also more capable of making informed buys.

This approach aims to reimagine the application of reviews in online shopping instead of merely limiting them. The system enhances accessibility, accuracy, and transparency by combining interactive chatbot delivery with AI-based review verification. Online shopping becomes more reliable and informed by integrating machine learning with real-time AI-based interaction.

In short, the proposed system presents a low-cost, flexible, and expandable solution to the problem of review authenticity verification, one of the most critical concerns of e-commerce. The approach fosters trust, transparency, and informed decision-making by creating a new norm for validating information in online commerce. By inducing ethical business behavior and genuine user experiences, the benefits extend to customers and vendors and strengthen the entire e-commerce ecosystem.

11. FUTURE WORK

Future advancements and scalability of the proposed system are very promising, with opportunities for improved review validation techniques, greater accessibility, and enhanced AI integration. The use of more advanced models than BERT, such as RoBERTa, T5, or GPT-based models, would significantly enhance the system's analytical power in the area of artificial intelligence. In addition, the inclusion of multimodal analysis—in which videos and images in reviews are also analyzed in addition to their text—would enhance validation processes and boost the accuracy of phoney review identification.

Biometric verification techniques, such as voice or facial identification, may also be employed to increase reviewer authentication even further and ensure that only authorised users join the review economy. Blockchain may also be an revolutionary addition. Immutable, tamper-proof verification of reviews would be possible through blockchain technology, enhancing the transparency and credibility of online commerce even more.

The multilingual nature of the system is an important area of development. Enabling the platform to be more inclusive and user-friendly for users from diverse linguistic and cultural backgrounds would be achievable through the inclusion of cross-lingual sentiment analysis, which would allow it to understand reviews in different languages. Additionally, to guarantee review authenticity across

markets, regional context-aware review validation could be employed to analyze cultural idiosyncrasies and shopping behaviors.

In order to efficiently manage larger sets of data, the system might be shifted to cloud-based infrastructure from a technical scaling perspective. Fault tolerance, modularity, and maintainability would all be enhanced by having a microservices-based architecture. Additionally, complete API development would enable seamless integration with third-party platforms, allowing e-commerce services to leverage the system for better review validation and fraud detection.

The reviewing process can be further defined by new user experience technologies such as voice-based reviewing and Augmented Reality (AR) for product visualisation. These would enhance how customers evaluate products prior to purchase by facilitating interactive user engagement with reviews.

With the incorporation of cutting-edge artificial intelligence (AI), blockchain security, cloud scalability, and immersive technologies, this system can potentially transform online review validation in the future and set new standards for user interaction, transparency, and trust.

12. REFERENCES

[1]. Punde, V., et al. (2021). "Fake Review Detection Using Sentimental Analysis". International Journal of Advanced Computer Science and Applications.

[2]. Mohawesh, R., Xu, S., Tran, S.N., Ollington, R., Springer, M., Jararweh, Y., & Maqsood, S. (2021). "Fake Reviews Detection: A Survey". IEEE Access, 9, 65771-65802.

[3]. Gayathri, M., Y.S.N Siva Teja, & K.Ajay Sharma (2023). "Fake Review Detection Using Machine Learning".

[4]. M Gayathri et al. (2023). "Fake Review Detection Using Machine Learning". International Journal of Information Technology.

[5]. Jindal and Liu (2018) developed a groundbreaking approach using linguistic patterns and behavioral features to identify fake reviews. (https://www.sciencedirect.com/journal/expert-systems-with-applications/vol/118).

[6]. Kumar et al. (2020) introduced an innovative time-series analysis method for detecting fake review patterns (https://ieeexplore.ieee.org/document/9123672).

[7]. Wang and Chen (2022) proposed an aspect-based sentiment analysis framework examining semantic consistency reviews (https://www.sciencedirect.com/science/article/abs/pii/S0957417422001985)

[8]. Rodriguez et al. (2023) explored Graph Convolutional Networks (GCN) integration with semantic similarity measures (https://www.mdpi.com/2076-3417/13/4/2244).

[9]. Singh and Patel (2023) developed a hybrid approach combining NLP with reviewer behavior analysis

(https://www.iieta.org/journals/ria/paper/10.18280/ria.370507)

[10]. Thompson et al. (2024) introduced a system leveraging transformer models to distinguish between AI-generated and human-written reviews (https://ieeexplore.ieee.org/document/9716933)